Percentages and ratios — Equivalents · Percentage of amounts · Interest · Ratio

A calculator is allowed for questions marked CALC. ★ = challenge question.

A · PERCENTAGES

1 Copy and complete the table of equivalents.

Fraction		$\frac{7}{20}$			$2\frac{2}{5}$
Decimal	0.6			1.25	
Percentage			15%		

$0.6 = \frac{3}{5} = 60\%$; $\frac{7}{20} = 0.35 = 35\%$; $15\% = 0.15 = \frac{3}{20}$; $1.25 = 1\frac{1}{4} = 125\%$; $2\frac{2}{5} = 2.4 = 240\%$.

2 Three groups of students tried a new snack. Group A: 80 of 200 liked it. Group B: 33 of 60. Group C: 63 of 140.

a Write each group's proportion who liked it as a percentage.

b Which group liked it most? Which group liked it least?

a A $\frac{80}{200} = 40\%$, B $\frac{33}{60} = 55\%$, C $\frac{63}{140} = 45\%$.
b Most: B. Least: A — percentages let you compare groups of different sizes.

3 A 750 ml fruit drink contains 300 ml of apple juice, 150 ml of mango juice and the rest is grape juice. What percentage of the drink is **a** apple, **b** mango, **c** grape?

a $\frac{300}{750} = 40\%$ **b** $\frac{150}{750} = 20\%$ **c** $100 - 40 - 20 = 40\%$ (300 ml).

4 A phone costs €480. In a sale it is reduced by 35%. Work out the sale price.

Multiplier for -35% is 0.65: Math input error. (Or 35% of 480 is 168; $480 - 168 = 312$.)

5 Omar gets a pay rise of 6%. His old salary was €25 000. What is his new salary?

Math input error.

6 **a** 10% of an amount is 4.8 kg. Work out the amount.

b 30% of an amount is 72 m. Work out the amount.

a $10\% \to 4.8$, so $100\% \to 48$ kg. **b** $30\% \to 72$, so $10\% \to 24$ and $100\% \to 240$ m.

7 CALC Lena invests €2400 for 3 years at 4.5% **simple** interest per year. How much is the investment worth after 3 years?

> Interest each year Math input error; over 3 years Math input error. Total Math input error.

8 CALC Kostas puts €3000 in an account paying 2% **compound** interest per year. How much is in the account after 3 years?

> Multiply by 1.02 three times: Math input error (to the nearest cent). Each year the interest is earned on a slightly bigger amount.

B · RATIOS

9 Simplify each ratio.

a $14 : 21$ **b** $2.5 : 7.5$ **c** $0.8 : 2$

> **a** $2 : 3$ ($\div 7$). **b** $1 : 3$ ($\div 2.5$). **c** $\times 10$ first: $8 : 20 = 2 : 5$.

10 Sophie mixes squash and water in the ratio $3 : 7$. She makes 450 ml of drink. How much squash and how much water does she use?

> $3 + 7 = 10$ parts; one part $= 450 \div 10 = 45$ ml. Squash $3 \times 45 = 135$ ml, water $7 \times 45 = 315$ ml (check: 450 ✓).

11 Green paint is made from blue, yellow and white in the ratio $5 : 3 : 2$. How much of each colour is needed for 2 litres of green paint?

> 10 parts; $2000 \div 10 = 200$ ml per part. Blue 1000 ml, yellow 600 ml, white 400 ml.

12 An alloy is 60% copper, 30% zinc and 10% nickel.

a Write the fraction of the alloy that is each metal, in simplest form.

b Write the ratio copper : zinc : nickel in its simplest form.

> **a** Copper $\frac{3}{5}$, zinc $\frac{3}{10}$, nickel $\frac{1}{10}$. **b** $60 : 30 : 10 = 6 : 3 : 1$.

13 A shop has 20% off everything. For one day it offers an extra 30% off the sale price.

a A jacket originally costs €150. Work out its price on that day.

b Is "20% off then 30% off" the same as 50% off? Explain.

> **a** $150 \times 0.8 = 120$, then Math input error.
>
> **b** No. The multipliers combine: $0.8 \times 0.7 = 0.56$, so you pay 56% — a 44% discount, not 50%. The second 30% is taken off a smaller amount.

14 After a 15% reduction, a pair of trainers costs €51. What was the original price?

> Math input error is 85% of the original: $1\% \to 0.60$, so Math input error. Check: $60 \times 0.85 = 51$ ✓

Check: percentage = part ÷ whole × 100 · multipliers: +6% → 1.06, –35% → 0.65 · simple interest is the same every year, compound grows · share in a ratio: find one part first.